

Static Cosmology

Lecture 6: The Static Distance Ladder

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Overview

Standard candles and standard rulers are combined after extinction correction to construct a static distance scale.

1 Observational setup

Source coordinates, observing band, selection function, and calibration state are fixed at the outset.

$$\mu = m - M = 5 \log_{10}(d/10pc) + A_v$$

2 Transport or equilibrium equation

The governing equation is solved first in a homogeneous limit and then with the stated spatial dependence.

$$\theta = L_{std}/d_A$$

3 Connection to data

Each model quantity is tied to a measured flux, spectrum, angle, count, or temperature.

$$p(d|D) \propto \prod_{\ell} L_{\ell}(d, \theta_m)$$

4 Degeneracies

Calibration, population, and medium parameters are varied separately to expose their degeneracies.

5 Example

The worked example carries units and uncertainty from the input catalog to the reported result.

Checks

- Verify dimensions and sign conventions.
- State the boundary conditions used in each limit.
- Report numerical precision separately from model uncertainty.

Reading

- Jeong, ch. 6
- Static Distance Ladder SDL-26
- Extinction Atlas §4